

Grocery List Optimizer

Project Report

Course Info: CSE215.10 | Faculty: SAM3 | Institution: NSU

Developer Information

❖ Muhtasim Naseef Rahil – 2611781642

Contribution: GroceryListOptimizer Class, Swing GUI,
Input-Output Management, File Handling.

❖ S M Shahriar Islam – 2531662642

Contribution: Perishable Interface, GroceryItem
and Item Category Classes

GitHub: <https://github.com/Hailstorm789/Grocerylistoptimiser>

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Abstract:

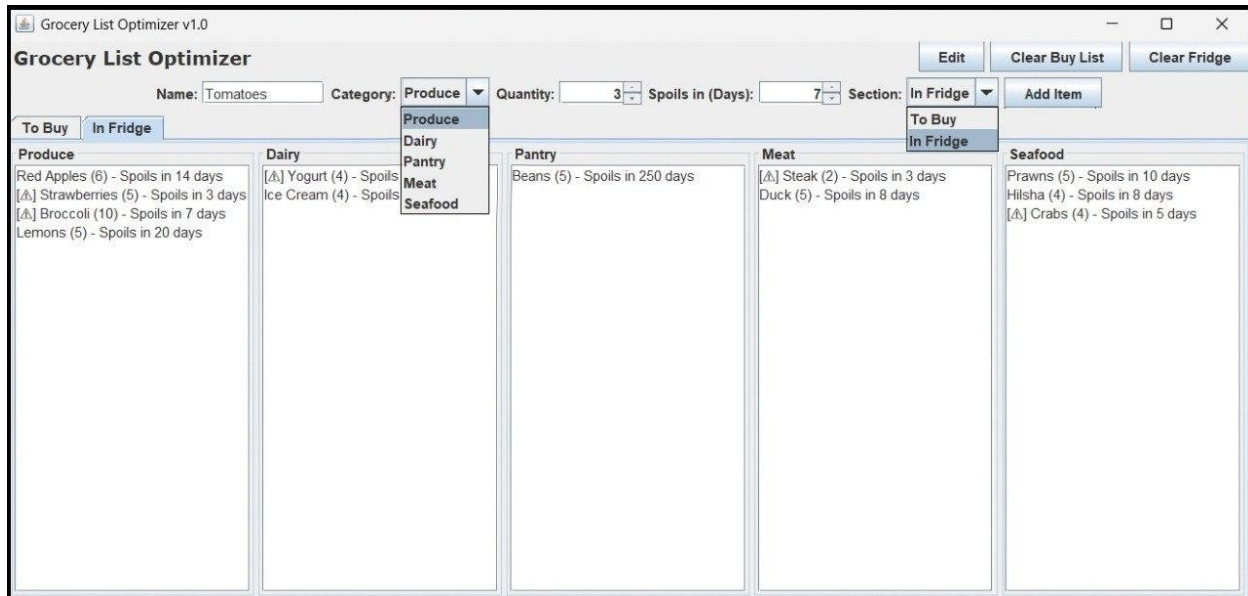
Grocery List Optimizer is a desktop application made using JAVA Swing, following the principles of Object Oriented Programming (OOP). The purpose of this application is to sort grocery items into dedicated categories based on item type such as Dairy, Meat, Pantry etc. as well as store quantity of those items. The application sorts items into dedicated To Buy or In Fridge tabs to keep track of which items need to be bought and which are in the fridge. It also detects the number of days until an item will be spoiled and marks those items with a warning sign, indicating that they need to be eaten quickly. All of these features combined create a useful application that can be used to keep track of everyday grocery items that are yet to be bought or already in the refrigerator and sort them accordingly in an organized list.

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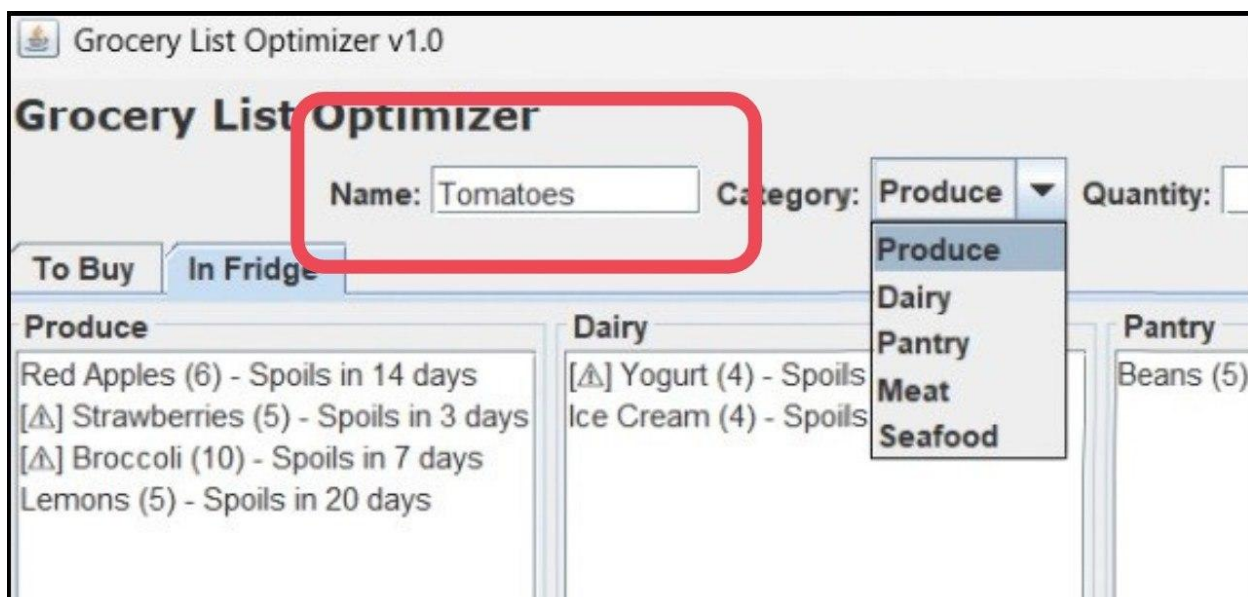
❖ Application GUI Overview:

The application shows an overview of your current to buy item lists as well as the items that are currently in the fridge. The top panel has all the input fields and formatting buttons to add/edit/remove items. The tabs can help you view to buy and in fridge section separately. Edit mode allows manual editing for further flexibility. The entire interface is made using JAVA Swing.



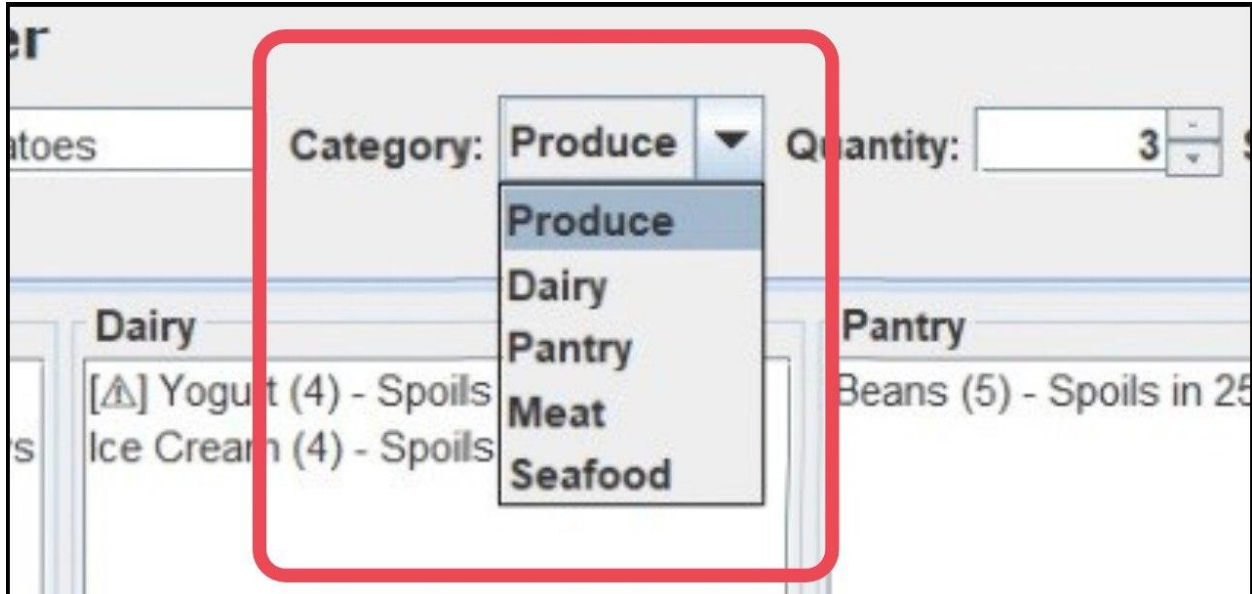
1. Name Field:

In the name field, you can enter the name of your item. It has been implemented using JTextField. Maximum name length can be changed in source code. By default, its 24 characters.



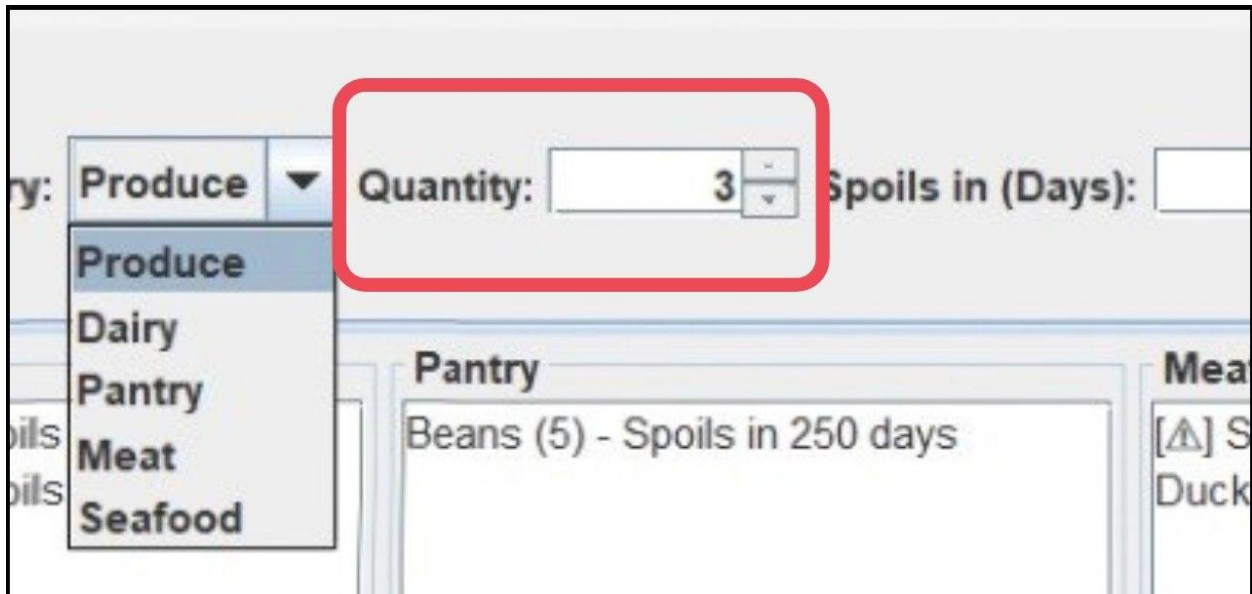
2. Category:

You can select item category from the category combo box. Currently available categories are Produce, Dairy, Pantry, Meat and Seafood.



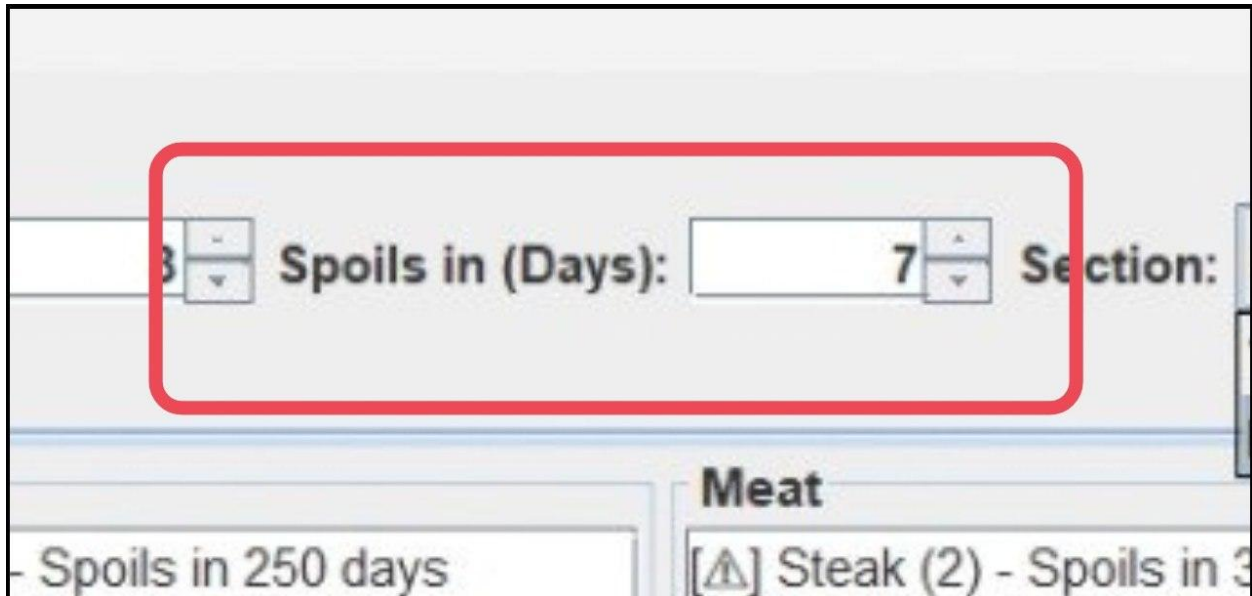
3. Quantity:

The quantity of item can be selected from the quantity spinner. The spinner ensures that given input stays within the hardcoded limit. Which is by default 1-1000.



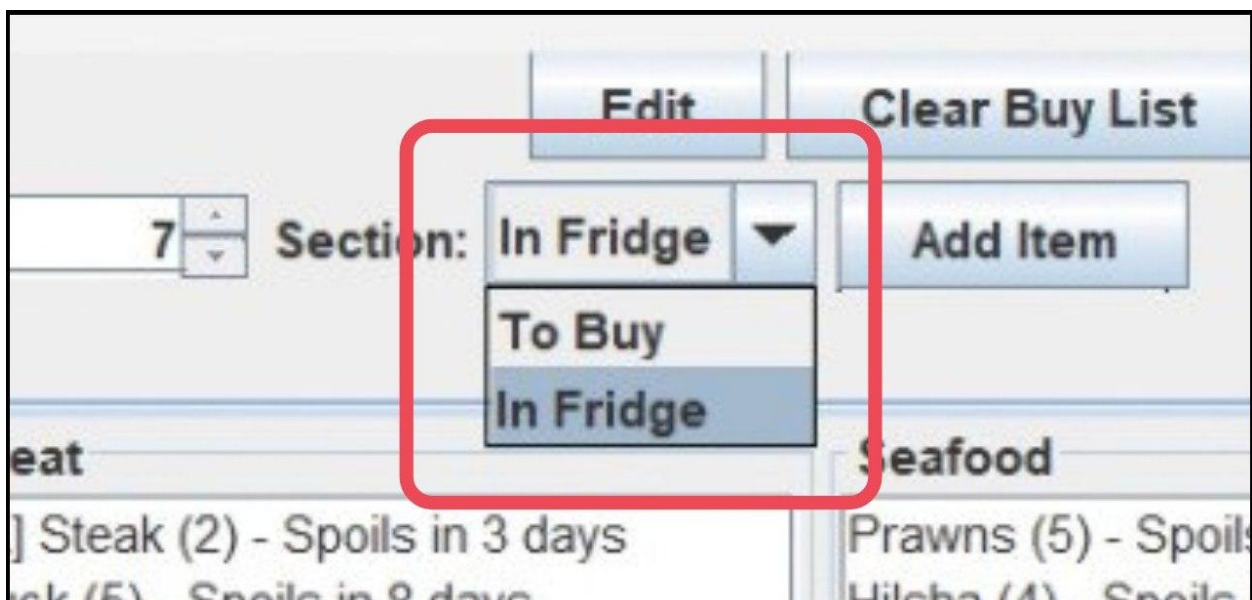
4. Spoiled in (Days):

The remaining days until an item will be spoiled can be set using the Spoils in (Days) spinner. The spinner makes sure that the input stays within hard coded limit which is 0-1461 days (4 year max)



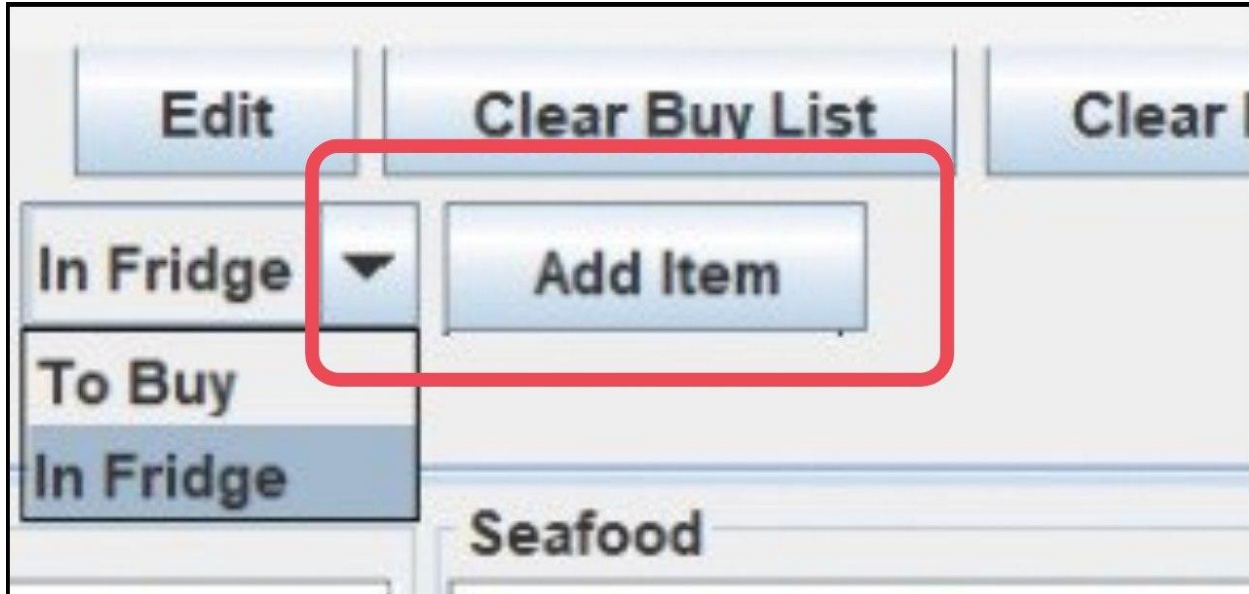
5. Section:

From the section combo box, you can list your items as To Buy or In Fridge Items.



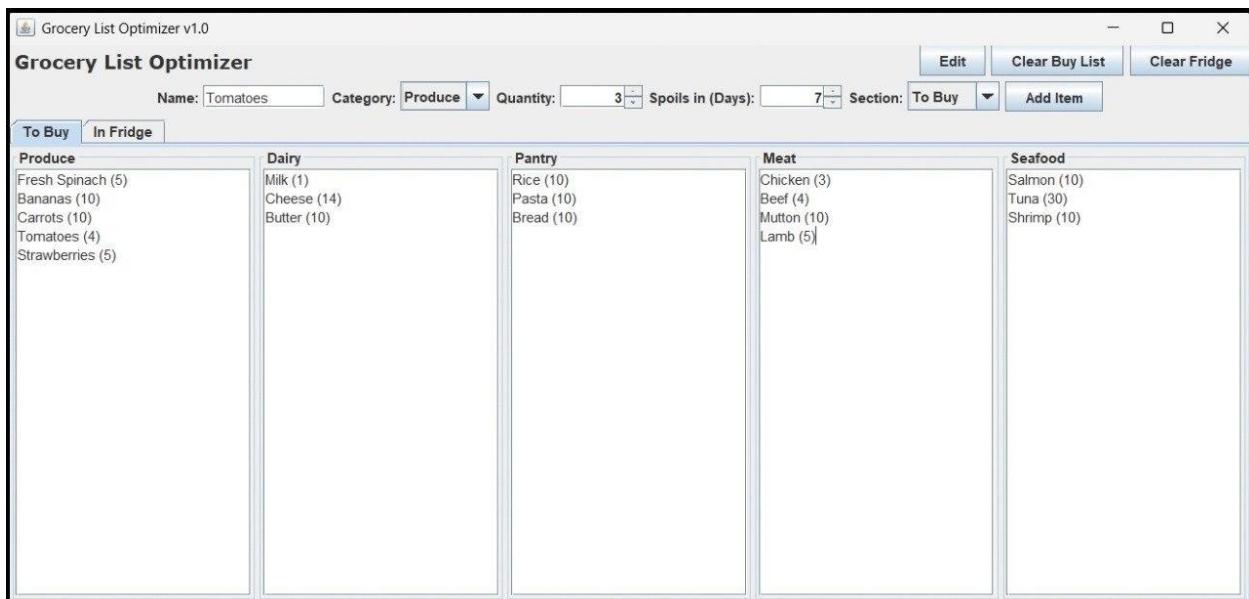
6. Add Item:

After providing all necessary inputs for a particular tab, clicking the Add Item button will create an item entry with all necessary information and store it on the section tab below.



7. To Buy Section:

The To Buy section shows the item names and quantities of items that needs to be bought.



8. In Fridge:

The In Fridge tab shows items that are in your fridge. It shows item names, quantities, and remaining days until an item will be spoiled.

The screenshot shows the 'Grocery List Optimizer v1.0' application window. The 'In Fridge' tab is selected. The interface includes a header with a title bar, a search bar, and several buttons: 'Edit', 'Clear Buy List', 'Clear Fridge', and 'Add Item'. Below the header, there are input fields for 'Name' (Tomatoes), 'Category' (Produce), 'Quantity' (3), and 'Spoils in (Days)' (7). The main content area is divided into five columns: Produce, Dairy, Pantry, Meat, and Seafood. Each column lists items with their quantities and spoilage dates. The 'In Fridge' tab is highlighted, and the 'Add Item' button is visible.

Produce	Dairy	Pantry	Meat	Seafood
Red Apples (6) - Spoils in 14 days	[⚠] Yogurt (4) - Spoils in 6 days	Beans (5) - Spoils in 250 days	[⚠] Steak (2) - Spoils in 3 days	Prawns (5) - Spoils in 10 days
[⚠] Strawberries (5) - Spoils in 3 days	Ice Cream (4) - Spoils in 14 days		Duck (5) - Spoils in 8 days	Hilsha (4) - Spoils in 8 days
[⚠] Broccoli (10) - Spoils in 7 days				[⚠] Crabs (4) - Spoils in 5 days
Lemons (5) - Spoils in 20 days				

9. Spoil Warning:

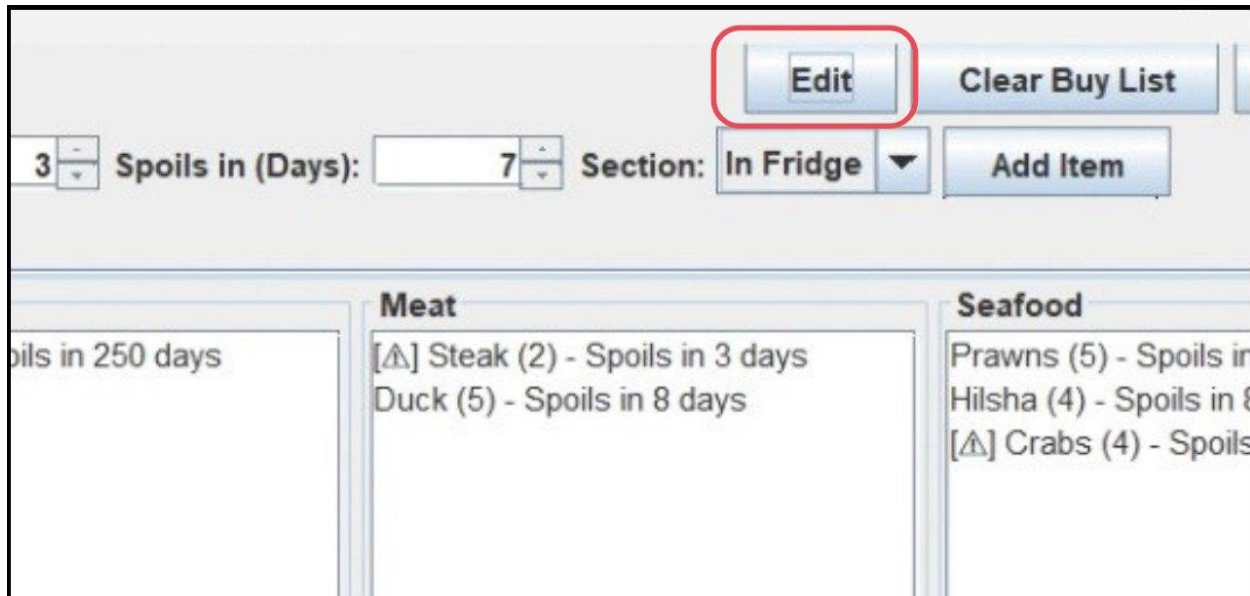
Items that will be spoiled in 7 days or less are highlighted with a warning sign, as an indication that this item needs to be eaten quickly. Maximum warning days can be edited in the source code.

This close-up screenshot focuses on the 'Dairy' section of the application. It shows two items: 'Yogurt (4) - Spoils in 6 days' and 'Ice Cream (4) - Spoils in 14 days'. The 'Yogurt' item is highlighted with a red box around the warning sign icon [⚠] and the text '6 days', indicating it is due for spoilage within 7 days. The 'Ice Cream' item is also highlighted with a red box around the text '14 days'. The 'Pantry' section is partially visible to the right, showing 'Beans (5) - Spoils in 250 days'.

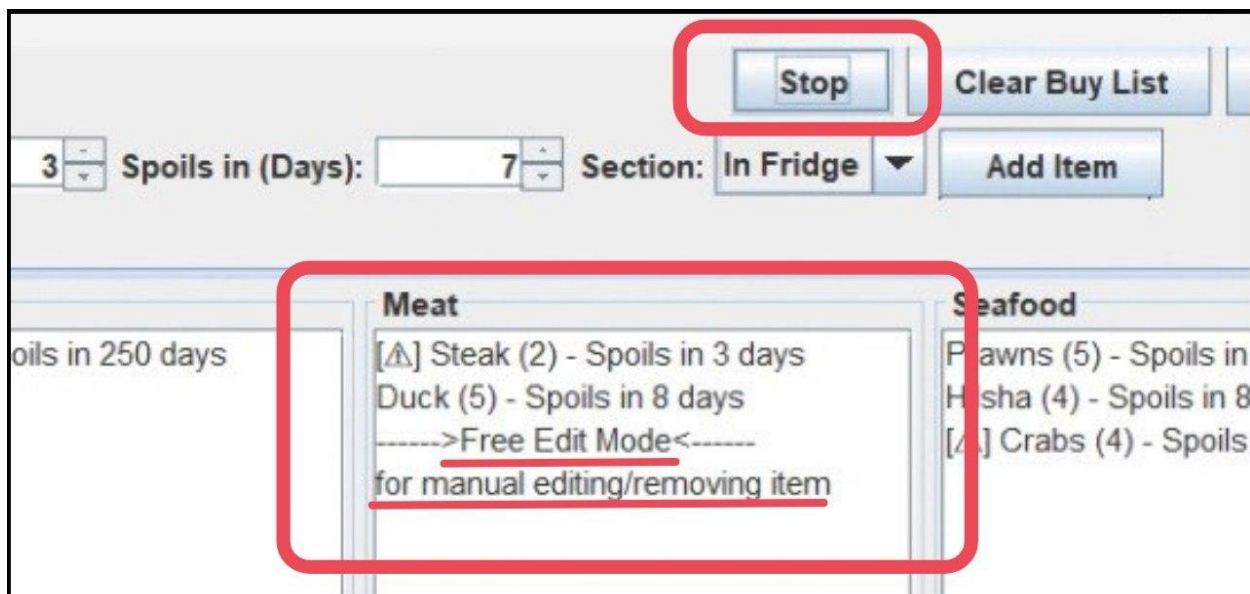
Dairy	Pantry
[⚠] Yogurt (4) - Spoils in 6 days	Beans (5) - Spoils in 250 days
Ice Cream (4) - Spoils in 14 days	

10. Edit Button:

The Edit button can be used to enter Free Edit mode. It basically allows you to manually change the information stored in the text areas below using your keyboard. You can add/edit/delete items freely. This allows further flexibility.



If you click on Stop button while you are in edit mode, the application will be set back to normal mode. The edited files will be saved even after closing the program.



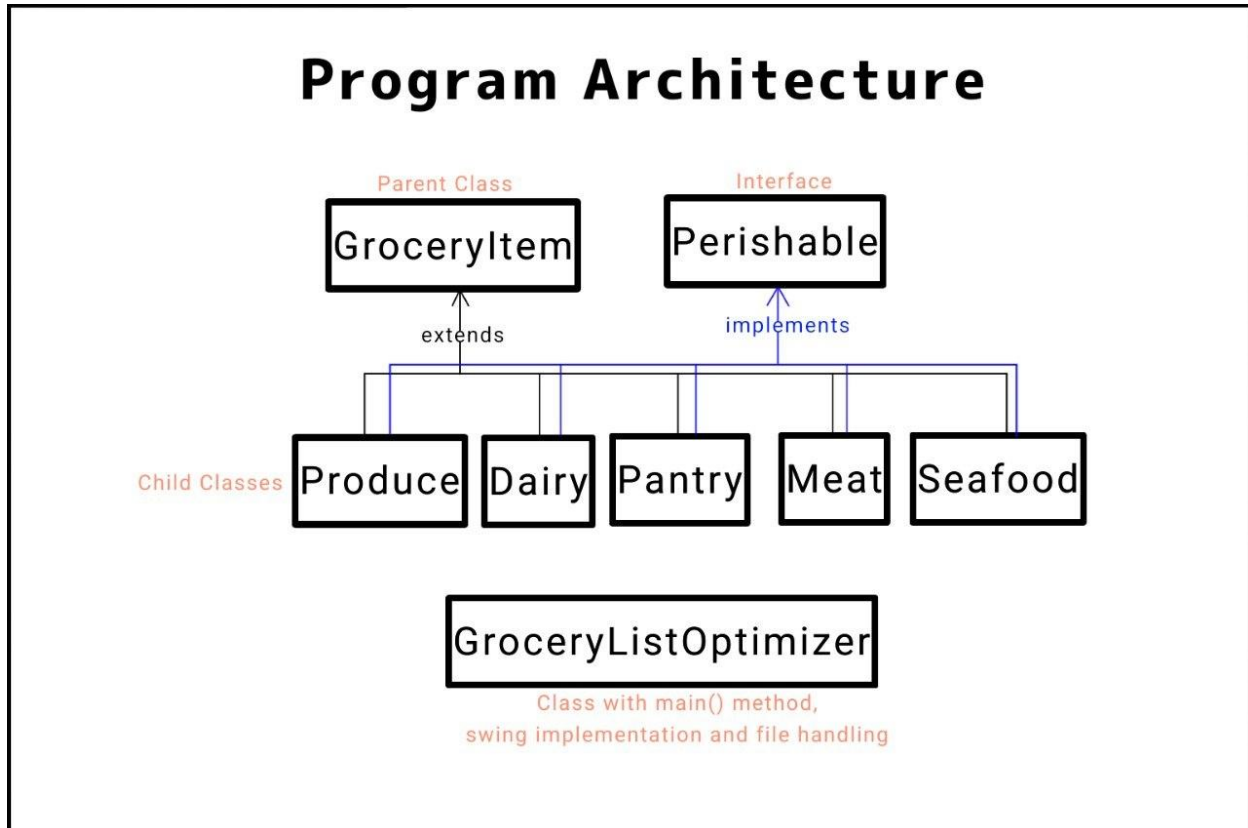
11. Clear Buttons:

Clicking the clear buttons will clear all information stored in a specific tab.

The screenshot displays the 'Grocery List Optimizer v1.0' application window. The title bar reads 'Grocery List Optimizer v1.0'. The main window has a title 'Grocery List Optimizer'. Below the title, there is a form with the following fields: 'Name: Tomatoes', 'Category: Produce' (with a dropdown arrow), 'Quantity: 3' (with a spinner), 'Spoils in (Days): 7' (with a spinner), and 'Section: In Fridge'. To the right of these fields are three buttons: 'Edit', 'Clear Buy List', and 'Clear Fridge'. The 'Clear Buy List' and 'Clear Fridge' buttons are highlighted with a red rectangular box. Below the form, there are two tabs: 'To Buy' and 'In Fridge'. The 'In Fridge' tab is selected. Below the tabs, there are five columns representing different categories: 'Produce', 'Dairy', 'Pantry', 'Meat', and 'Seafood'. Each column is currently empty.

❖ OOP Architecture Overview:

The entire programmed is made following the principles of Object Oriented Programming. Here is a comprehensive demonstration to explain the core architecture of the program:



1. GroceryItem (Abstract Parent Class)

The **GroceryItem** class is an abstract class which includes name and quantity variables and dedicated getter methods to access those files. It includes abstract methods such as `getLocation()`, `getAisleCategory()` and `getInfo()`. It is also the Parent Class of item type classes and its abstract methods have been overridden in the child classes.

2. Perishable (Interface)

The **perishable** interface includes `getDaysUntilSpoiled()` method. This interface has been implemented by item classes that can be spoiled and those classes have overridden the method.

3. Produce, Dairy, Pantry, Meat, Seafood (Child Classes)

The item type classes *extends* the GroceryItem class and *implements* the Perishable interface. These classes have additional type, daysUntilSpoiled and inFridge Boolean variable. They override all abstract methods from GroceryItem Class and Perishable Interface. Methods such as getLocation() to get tab location, getAisleCategory() to get category of an item object, getInfo() to get necessary information regarding an item object and getDaysUntilSpoiled to remaining days for an item are present.

4. GroceryListOptimizer Class

This class includes the main method, JAVA Swing GUI, Polymorphic Object Creation and file handling. This class utilizes all other classes and uses its own methods to make the program work.

❖ Functional Overview

Most of the program's main functionalities such as global constant variables, GUI features, input management, polymorphic object creation and file handling are inside GroceryListOptimizer.java file. This file utilizes all the other classes and interface to make the program work.

1. Global Variables

The global variables are declared on top of the program. Their values can be edited to change Category and Section Tab names, Maximum Name Length, Maximum Limit for Quantity, Days Until Spoiled and Warning Days. Furthermore, Input fields and output text areas are also added as global variables so that it can be accessed from other methods of the class.

2. GUI Architecture

The main() method includes Java swing implementation to create the full GUI of the program. Different panels have been used to combine certain portions of the GUI and then those panels were combined into the main frame to build the GUI. BorderLayout class has been used to position the elements and the panels in an organized way.

User Interface Construction:

Upper Panel: Title Text, Format Buttons Panel

Format Buttons Panel: Edit, Clear Buy List and Clear Fridge Buttons

Input Panel: Name TextField, Category & Section Combobox,

Quantity & Spoiled (Days) Spinner, Add Item Button

Top Panel: Upper Panel + Input Panel

Section Tabs: To Buy & In Fridge Tab

Main Frame: Top Panel + Section Tabs

3. Polymorphic Item Object Creation

After filling necessary input fields and clicking the Add Item button, the `addItem()` method evaluates the given input and based on the selected category, it uses polymorphism to create the appropriate child-class object. After that, it saves the items using `saveItem()` method, resets input fields back to default using `resetInputs()` and loads newly updated data using `loadData()` method.

4. File Handling

`saveItem()` Method & Warning Validation: Saves Items into dedicated files based on category and section. It also validates *warning sign* based on remaining days until the item will be spoiled. It writes the files using *append* mode to ensure the files are created if they don't exist and are updated if they already exist.

`loadData()` Method: Loads current file data into text areas with the help of `readFile()` method.

`readFile()` Method: Reads the files and appends information to dedicated text areas in GUI.

`toggleEditable()` Method: Toggles text areas `isEditable` Boolean variable, allowing free text editing.

`saveCurrentText()` Method: Saves edited text before stopping edit mode with the help of `writeTextToFile()` method.

`writeTextToFile()` Method: Writes edited from text areas into dedicated files to update them.

❖ Conclusion:

In conclusion, *Grocery List Optimizer* successfully solves the problem of inefficient super market shopping with the help of a clean, compact graphical user interface. Furthermore, it keeps track of currently stored food items in the fridge and warn users to eat them before they get spoiled. Additionally, it provides user-freedom to edit the list freely for further usability. All of this features combined creates a very useful application that perfectly compliments everyday grocery shopping.